

Decent Coaching Classes

X Semi

Mind Your Math - VII

Answer Key & Marking Scheme

Marks: 30

Time: 1 Hr

Q.1 (A) Correct alternatives

[4]

- (1) (A) -3 ($D = 2 \times (-1) - 1 \times 1 = -3$) [1]
- (2) (A) 2 and 3 [1]
- (3) (B) -3 ($7 - 10 = -3$) [1]
- (4) (B) 19 ($1 + 9 \times 2$) [1]

Q.1 (B) Solutions

[4]

- (1) $3x^2 - 5x + 2 = 0$ [1]
- (2) $a = 1, b = -3, c = 2; b^2 - 4ac = 9 - 8 = 1$ [1]
- (3) No. Differences: $4 - 1 = 3, 9 - 4 = 5, 16 - 9 = 7$ are not equal, so there is no common difference. [1]
- (4) 7, 10, 13 [1]

Q.2 (A) Any two (2 marks each)

[4]

- (1) $D = -1 - 2 = -3; D_x = -4 - 2 = -6; D_y = 2 - 8 = -6$ [1]
 $x = -6/-3 = 2; y = -6/-3 = 2$ [1]
- (2) $x^2 - 3x - 4x + 12 = 0$ [$\frac{1}{2}$]
 $x(x - 3) - 4(x - 3) = 0$ [$\frac{1}{2}$]
 $(x - 3)(x - 4) = 0, x = 3$ or $x = 4$ [1]
- (3) $a = 12, d = 4$ [$\frac{1}{2}$]
 $t_{10} = 12 + (10 - 1) \times 4 = 12 + 36$ [1]
 $= 48$ [$\frac{1}{2}$]

Q.2 (B) Any two (2 marks each)

[4]

- (1) Adding the two equations: $3x = 12$, so $x = 4$ [1]
 $y = 7 - 4 = 3. (x, y) = (4, 3)$ [1]
- (2) $a = 1, b = -6, c = 9. b^2 - 4ac = 36 - 36 = 0$ [1]
Discriminant is zero, so the roots are real and equal. [1]
- (3) $a = 3, d = 5, n = 12. t_n = a + (n - 1)d$ [1]
 $t_{12} = 3 + 11 \times 5 = 3 + 55 = 58$ [1]

Q.3 (A) Any one (3 marks)

[3]

- (1) $b = 10, c = 9; b^2 - 4ac = 100 - 36 = 64$ [1]
 $x = (-10 \pm 8)/2$ [1]
 $x = -1$ or $x = -9$ [1]
- (2) $d = 5$ [1]
 $79 = 4 + (n - 1) \times 5, 75 = 5(n - 1), n - 1 = 15$ [1]
 $n = 16. 79$ is the 16th term. [1]

Q.3 (B) Any two (3 marks each)

[6]

- (1) $D = 3 \times 2 - 1 \times 1 = 5$ [1]

$$D_x = 7 \times 2 - 1 \times 4 = 10; D_y = 3 \times 4 - 7 \times 1 = 5 \quad [1]$$

$$x = 10/5 = 2; y = 5/5 = 1 \quad [1]$$

(2) $2 \times (-3) = -6$; factors of -6 with sum 5 are 6 and -1 [1]

$$2x^2 + 6x - x - 3 = 0, 2x(x + 3) - 1(x + 3) = 0, (x + 3)(2x - 1) = 0 \quad [1]$$

$$x = -3 \text{ or } x = 1/2 \quad [1]$$

(3) $a = 7, d = 4, n = 20$ [1]

$$t_{20} = 7 + (20 - 1) \times 4 = 7 + 76 \quad [1]$$

$$= 83 \quad [1]$$

Q.4 Any one (5 marks)

[5]

(1) Let cost of a book = ₹ x , cost of a pen = ₹ y . $2x + 3y = 90$... (i); $3x + 2y = 110$... (ii) [1]

(i) $\times 2$: $4x + 6y = 180$; (ii) $\times 3$: $9x + 6y = 330$ [1]

Subtracting: $5x = 150$, so $x = 30$ [1]

From (i): $60 + 3y = 90$, so $y = 10$ [1]

A book costs ₹30 and a pen costs ₹10. [1]

(2) Let the integers be x and $x + 1$. $x(x + 1) = 132$ [1]

$$x^2 + x - 132 = 0 \quad [1]$$

$$x^2 + 12x - 11x - 132 = 0, \text{ so } (x + 12)(x - 11) = 0 \quad [1]$$

$$x = -12 \text{ (rejected, not positive) or } x = 11 \quad [1]$$

The integers are 11 and 12. [1]

(3) The savings 100, 150, 200, ... form an A.P.: $a = 100, d = 50$ [1]

(i) $t_{10} = 100 + (10 - 1) \times 50$ [1]

$$= 100 + 450 = ₹550 \quad [1]$$

(ii) $800 = 100 + (n - 1) \times 50$, so $700 = 50(n - 1)$, $n - 1 = 14$ [1]

$$n = 15. \text{ She saves ₹800 in the 15th month. } [1]$$